



University College Dublin
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MID-MODULE TEST, 1 DEIREADH FÓMHAIR/OCTOBER 2026

COMP 10280
Programming I (Conversion)

Module Co-ordinator: Professor John Dunnion

Time allowed: 1 hour.

Instructions for Candidates

Answer both questions.

Students should write programs in Python 3.
Pseudocode for algorithms should be included with programs.

Instructions for Invigilators

Students only have access to:

- Brightspace;
- lecture slides for the module;
- the official Python Website, <http://www.python.org>; and
- a translation Webpage (eg Google Translate).

No other Internet access is permitted.

1. Write a program that prompts the user for a positive number (ie an integer > 0) and uses a while loop to calculate the number of integers from 1 up to and including that number that are evenly divisible by 5, 7, 13, 17 and 19. The program should print out an error message if a negative number or 0 is entered.

Sample outputs from this program are as follows (each output represents a separate execution of the program, ie the program prompts for a single number, processes it and exits):

Program to calculate the number of integers evenly divisible by 5, 7, 13, 17 and 19

Enter a positive integer: 18

You entered: 18

Number of numbers from 0 up to and including 18 evenly divisible by 5: 3

Number of numbers from 0 up to and including 18 evenly divisible by 7: 2

Number of numbers from 0 up to and including 18 evenly divisible by 13: 1

Number of numbers from 0 up to and including 18 evenly divisible by 17: 1

Number of numbers from 0 up to and including 18 evenly divisible by 19: 0

Finished!

Program to calculate the number of integers evenly divisible by 5, 7, 13, 17 and 19

Enter a positive integer: 200

You entered: 200

Number of numbers from 0 up to and including 200 evenly divisible by 5: 40

Number of numbers from 0 up to and including 200 evenly divisible by 7: 28

Number of numbers from 0 up to and including 200 evenly divisible by 13: 15

Number of numbers from 0 up to and including 200 evenly divisible by 17: 11

Number of numbers from 0 up to and including 200 evenly divisible by 19: 10

Finished!

Program to calculate the number of integers evenly divisible by 5, 7, 13, 17 and 19

Enter a positive integer: 0

You entered: 0

Number entered should be > 0 .

Finished!

Program to calculate the number of integers evenly divisible by 5, 7, 13, 17 and 19

Enter a positive integer: -200
You entered: -200
Number entered should be > 0.
Finished!

Program to calculate the number of integers evenly divisible by 5, 7, 13, 17 and 19

Enter a positive integer: 20
You entered: 20
Number of numbers from 0 up to and including 20 evenly divisible by 5: 4
Number of numbers from 0 up to and including 20 evenly divisible by 7: 2
Number of numbers from 0 up to and including 20 evenly divisible by 13: 1
Number of numbers from 0 up to and including 20 evenly divisible by 17: 1
Number of numbers from 0 up to and including 20 evenly divisible by 19: 1
Finished!

Program to calculate the number of integers evenly divisible by 5, 7, 13, 17 and 19

Enter a positive integer: 10000
You entered: 10000
Number of numbers from 0 up to and including 10000 evenly divisible by 5: 2000
Number of numbers from 0 up to and including 10000 evenly divisible by 7: 1428
Number of numbers from 0 up to and including 10000 evenly divisible by 13: 769
Number of numbers from 0 up to and including 10000 evenly divisible by 17: 588
Number of numbers from 0 up to and including 10000 evenly divisible by 19: 526
Finished!

[15 marks]

2. The 2026–2027 academic year in a particular Indian university runs from 1 August 2026 to the end of July the following year. Write a program that prompts the user for a series of days (integer numbers) and, for each day, prints out the date, the month and the year in the 2026–2027 academic year of that day. The program should continue until a non-positive number is entered.

Sample output from this program is as follows:

Program to calculate the date, month and year in an Indian academic year 2026-2027 of a given day.

Enter 0 or a negative number to exit.

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Enter the day for which you want to find the date (a positive integer): 1
You entered: 1
Day number 1 is 1 August 2026
Enter the day for which you want to find the date (a positive integer): 31
You entered: 31
Day number 31 is 31 August 2026
Enter the day for which you want to find the date (a positive integer): 32
You entered: 32
Day number 32 is 1 September 2026
Enter the day for which you want to find the date (a positive integer): 62
You entered: 62
Day number 62 is 1 October 2026
Enter the day for which you want to find the date (a positive integer): 147
You entered: 147
Day number 147 is 25 December 2026
Enter the day for which you want to find the date (a positive integer): 154
You entered: 154
Day number 154 is 1 January 2027
Enter the day for which you want to find the date (a positive integer): 365
You entered: 365
Day number 365 is 31 July 2027
Enter the day for which you want to find the date (a positive integer): 366
You entered: 366
Day number 366 is not in the Indian academic year 2026-2027!
Enter the day for which you want to find the date (a positive integer): 183
You entered: 183
Day number 183 is 30 January 2027
Enter the day for which you want to find the date (a positive integer): 0
Finished!
```

(For the removal of doubt, there are 30 days in each of the months of April, June, September and November, there are 28 days in February 2027 and there are 31 days in each of the other months of the year.)

[35 marks]